Ashish Kumar Nanda

(+91) 9279741216 — ashish.nanda1902@gmail.com — linkedin.com/ashishnanda19 — github.com/ashishnanda19 — Portfolio — leetcode.com/ashishnanda19 — codechef.com/ashishnanda19

Summary

Full-stack developer who builds scalable, event-driven backend systems, microservices, real-time applications, and AI-powered products end to end, from API design to deployment. Strengths include designing clean, reliable APIs, turning product ideas into production-ready software, and troubleshooting complex systems. Also experienced in building LLM-based applications and applying optimization research to real-world problems, backed by strong data structures and algorithms fundamentals.

Education

Manipal University Jaipur **Aug 2023 - Jul 2027**

Bachelors of Technology in Computer Science and Engineering *CGPA: 9.12*

Core: Operating Systems, Data Structures & Algorithms, OOP, DBMS, Software Engineering, Computer Networks

Professional Experience

Indian Institute of Technology (IIT) BHU **Dec 2025 - May 2026**

Research Intern *Varanasi, Uttar Pradesh*

- Developed a **MILP** optimization model in **Gurobi** for MSME supply chain, achieving **5.1% lower cost** and **33% lower emission** through optimized production, energy, transport, and technology decisions.
- Modeled **4+ green technologies** across **3 MSME categories**, integrating government subsidies to identify cost-effective technology adoption and scalable **decarbonization strategies**.

Projects

VidFlow - Distributed Video Transcoding Platform [Live] [GitHub]

Node.js, Express.js, AWS(S3, Lambda, ECS), MongoDB, Redis, Docker, ffmpeg, CUDA

- Built a **distributed, event-driven video transcoding platform** that automatically generates multi-resolution outputs **(360p - 1080p)** with **leaky bucket rate-limiting** and a **Redis job queue** for efficient concurrency.
- Utilized AWS **S3, Lambda and ECS** for seamless event handling, container orchestration, and scalable backend operations with **signed URLs** for secure media access control.
- Added an **NVIDIA CUDA (NVDEC/NVENC)** transcoding path with automatic **CPU fallback**, and a single-pass encode that cut CPU transcode time from **27s to 14s (1.9x faster)**.

SafeTrail - Emergency Response Platform [Live] [GitHub]

Node.js, Express.js, AWS(S3, RDS, ECS), Socket.IO, PostgreSQL, Redis, BullMQ

- Architected a real-time **emergency response platform** with live location sharing at **under 200ms latency** and **async SOS workflows** via Redis, maintaining **API response under 150ms** during concurrent usage.
- Implemented a **geospatial intelligence system** using **PostGIS** to identify nearest services within a **50km radius in under 50ms with 70%** faster lookups, and integrated **voice-triggered SOS detection** with confidence thresholds.

HyperRAG-X - Enterprise Hybrid RAG Platform [Demo] [GitHub]

Python, FastAPI, LangChain, LangGraph, Groq, Qdrant, NetworkX, React, Supabase

- Designed a **4-agent LangGraph pipeline** with **hybrid retrieval** across **Qdrant**, **NetworkX**, and memory cache, enabling multi-hop reasoning with **70B parameter** LLM inference via Groq.
- Developed an async **Macrophage Agent** for continuous LLM-based contradiction detection and shipped a full-stack product with **FastAPI**, **Supabase** auth, and a React-based interface with live **force-graph** knowledge visualization.

Skills

Languages: JavaScript, Python, C++, Java, SQL
Backend & APIs: Node.js, Express.js, RESTful APIs, FastAPI, Socket.IO
Frontend: React.js, Tailwind CSS, HTML, CSS
Databases: MySQL, PostgreSQL, MongoDB, Redis, ChromaDB
LLM & GenAI: LangChain, LangGraph, Hugging Face, RAG, Llama, Groq
Cloud & Tools: Git, GitHub, Postman, AWS, Docker, CUDA, CI/CD, Supabase

Achievements

- **Competitive Programming** – LeetCode: **750+** solved, peak rating **1,832 (Top 6.68%)**; CodeChef: **3 star** (max **1645**), Global Rank **311** & **256** in Starters 247 & 260; Codeforces: **Pupil** (max **1210**).
- **National Semifinalist - Flipkart GRiD 7.0** Ranked among 160,000 participants nationwide. [Certificate]
- **Finalist - International Innovation Challenge (IIC) 2.0** Among 1,100+ participants across India. [Certificate]
- **4X Dean's List of Excellence:** Recognized for academic excellence. [Certificate]